Training-Free Token Reduction on MotionBench
Results

HuVLLM Project

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Eleven training-free video-LLM token-reduction methods, evaluated on MotionBench (8,052 questions, 4,018 scoreable) across two backbones — LLaVA-OV-7B and Qwen3-VL-8B — at 32 frames. Results only: full methodology, the verification gate, and limitations are in `paper.tex`. At $n = 4{,}018$ the binomial 95% confidence half-width is ± 1.54 points; * marks a difference significant by McNemar's test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$).

Contents

1 Common settings

Held fixed across every method and both backbones:

- **32 frames**, sampled uniformly per video.
- **Greedy decoding** (`do_sample=False`, `max_new_tokens=16`), batch size 1.
- **Letter-match scoring** (A–D) against ground truth; the 4,034 `NA` items are excluded, leaving 4,018 scoreable.
- Methods exposing a single retention knob are standardized to **15% nominal retention**; methods with no such knob (frame-selection methods, or methods whose budget is fixed internally) keep their published defaults — forcing 15% on those would break paper fidelity.

Per-method settings, mechanisms and measured token budgets are in Table 5 (§5).

2 Master table

Table 1 is every method at its standardized or default setting, both backbones, all six subcategories, both baselines. Four of eleven methods expose a retention knob; PruneVID's authors publish exactly one operating point (`cluster_ratio=0.5`, †), so its row here is that 50% cell, not a 15% run. The other three — FastV, FlashVID, HoliTom — have a knob the authors validate at multiple points; that data is Table 2, immediately below. § AIM's released code ships two active merge steps, so its LLaVA-OV cell runs at 25% retention, not 15%. Rounding is half away from zero throughout (see note, Table 1 caption).

Table 1: Master results table: every method at its standardized or default setting, grouped by backbone (no separate Backbone column — see the section divider rows). AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count (% correct within category, chance = 25%). Δ is vs. that row's backbone baseline. * = significant. † PruneVID's only published operating point is `cluster_ratio=0.5`; this row is that 50% cell, not a 15% run — it was never run at 15% on either backbone. Six Repetition Count cells land on an exact x.25 rounding tie (e.g. 101/400); this table rounds half away from zero, matching `paper.tex` — naive recomputation with default rounding disagrees on these cells by 0.1 point.

Method	Retention	Overall	Δ	AO	CM	LM	MR	MO	RC
LLaVA-OV-7B, standardized setting (baseline 52.66)									
Baseline	—	*52.66*	—	40.5	45.2	55.5	57.0	71.2	23.8
DyCoke	15%	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	15%	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	15%	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	15%	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	15%	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM	25%§	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
STTM	15%	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV	50%†	38.20	–14.46*	32.0	34.8	35.5	38.4	52.9	27.3
FastV	15%	36.78	–15.88*	32.8	31.9	34.1	35.7	53.8	25.3
Qwen3-VL-8B, standardized setting (baseline 62.52)									
Baseline	—	*62.52*	—	46.1	63.1	65.0	67.3	79.0	33.8
PruneVID	50%†	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	15%	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM	15%	61.60	–0.92*	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom	15%	60.33	–2.19*	44.7	61.3	61.0	66.4	76.2	29.0
MDP3	15%	59.66	–2.86*	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID	15%	59.36	–3.16*	43.5	57.4	62.6	65.6	77.2	23.5
FastV	15%	59.01	–3.51*	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (contextual)	15%	58.81	–3.71*	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG	15%	56.35	–6.17*	45.3	53.5	59.3	60.2	75.4	22.2
AIM	15%	55.97	–6.55*	45.9	56.6	59.5	58.3	72.2	27.0

Table 1 fixes every method to its standardized or default setting; three methods (FastV, FlashVID, HoliTom) have an authors'-validated retention knob and were run at every point the authors themselves publish. Table 2 is that data on its own, ordered 10/15/20/25% per backbone rather than split across the rows above.

Table 2: The three methods with an authors'-validated multi-point retention sweep, at every published point. * = significant vs. that backbone's baseline. — = not run: 20% was never run on Qwen3-VL for any method, and FastV's LLaVA-OV range already extends to 50/75% (Table 5 note), making a 20% point there redundant. ‡ FlashVID's Qwen3-VL 10%/25% cells predate the authors'-code fix (§5): the incomplete reimplementation, not the verified method. Span is 10%→25% movement; not reported for FlashVID/Qwen3-VL since neither endpoint is the verified method. PruneVID is not in this table — its authors publish exactly one operating point (Table 1's † note), not a sweep.

Method	$r = 0.10$	$r = 0.15$	$r = 0.20$	$r = 0.25$	Span
LLaVA-OV-7B					
Baseline		*52.66*			—
FastV	36.06*	36.78*	—	36.73*	+0.67
FlashVID	52.51	53.31	53.71*	53.36	+0.85
HoliTom	52.76	53.14	53.06	53.41	+0.65
Qwen3-VL-8B					
Baseline		*62.52*			—
FastV	56.92*	59.01*	—	60.60*	**+3.68**
FlashVID‡	54.73*	59.36*	—	59.36*	—
HoliTom	60.05*	60.33*	—	60.43*	+0.38

FlashVID's 20% LLaVA-OV cell is the one retention point in this report that crosses significance in the direction of a gain ($\chi^2 = 3.91$, 236 fixed vs. 194 broke); HoliTom's does not ($\chi^2 = 0.43$). Both gate clean. It survives at only one of four points, so it is reported because it crosses the threshold, not because it changes the picture that FlashVID is accuracy-neutral on LLaVA-OV. FastV's own published range extends past this table to 50% and 75% keep (both land at 36.3–36.4%, the same band as every other point); see Table 5.

FlashVID's two Qwen3-VL reimplementation cells (10%, 25%) show 54.73% and 59.36% — the latter identical to the verified 15% cell above, to two decimal places. This is coincidence, not a duplicated run: the two share only 5,725 of 8,052 predictions, confirmed independently.

3 Results by backbone

3.1 LLaVA-OV-7B

Table 3: LLaVA-OV-7B, 32 frames, 15% nominal retention except where noted. All nine methods that run on this backbone. § AIM runs at 25% retention (two active merge steps ship in the released code). † PruneVID-OV runs at `cluster_ratio=0.5` (50% retention) — the authors' only published operating point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM§	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
Baseline	*52.66*	—	*40.5*	*45.2*	*55.5*	*57.0*	*71.2*	*23.8*
STTM	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV*†	38.20	–14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	–15.88	32.8	31.9	34.1	35.7	53.8	25.3

3.2 Qwen3-VL-8B

Table 4: Qwen3-VL-8B, 32 frames, 15% nominal retention except where noted. All ten portable methods. VisionZip is the contextual half only — Qwen3-VL's vision tower has no CLS token, so the dominant half is not implementable; not the published method. FlashVID and STTM run the authors' own code at published settings (§5); neither carries the off-spec caveat from earlier revisions of this report. † PruneVID runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*62.52*	—	*46.1*	*63.1*	*65.0*	*67.3*	*79.0*	*33.8*
PruneVID†	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM*	61.60	–0.92	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom*	60.33	–2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	–2.86	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID*	59.36	–3.16	43.5	57.4	62.6	65.6	77.2	23.5
FastV*	59.01	–3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	–3.71	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG*	56.35	–6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	–6.55	45.9	56.6	59.5	58.3	72.2	27.0

4 Figures

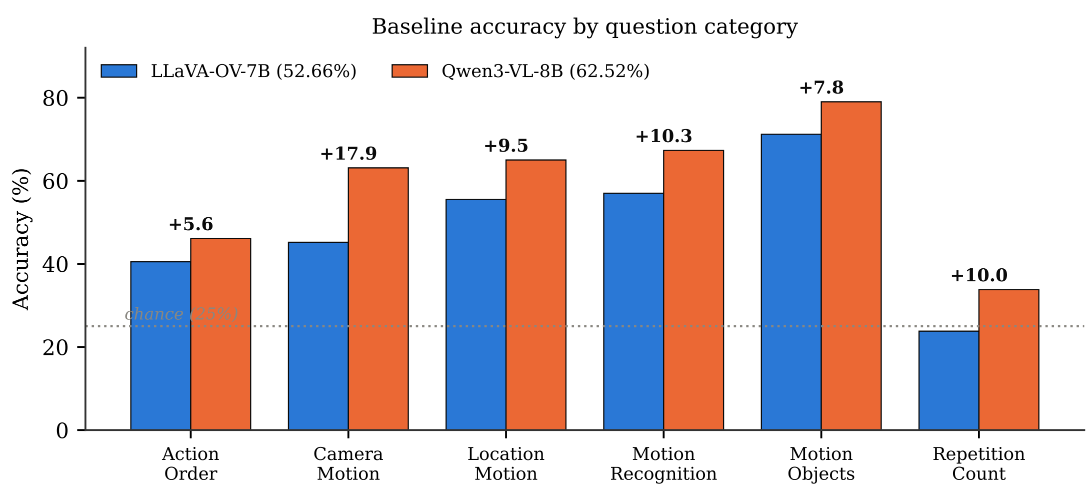

Figure 1: Baseline accuracy by category, both backbones (bar).

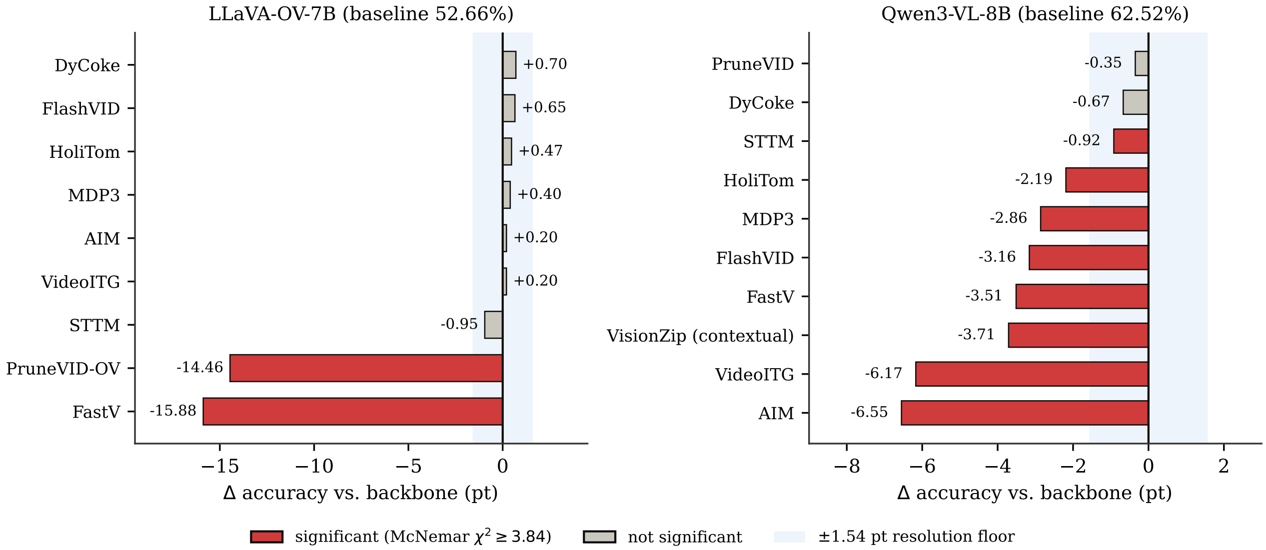

Figure 2: Change vs. backbone, every method, both backbones (bar). Red = significant by McNemar's test; the shaded band is the ±1.54-point resolution floor.

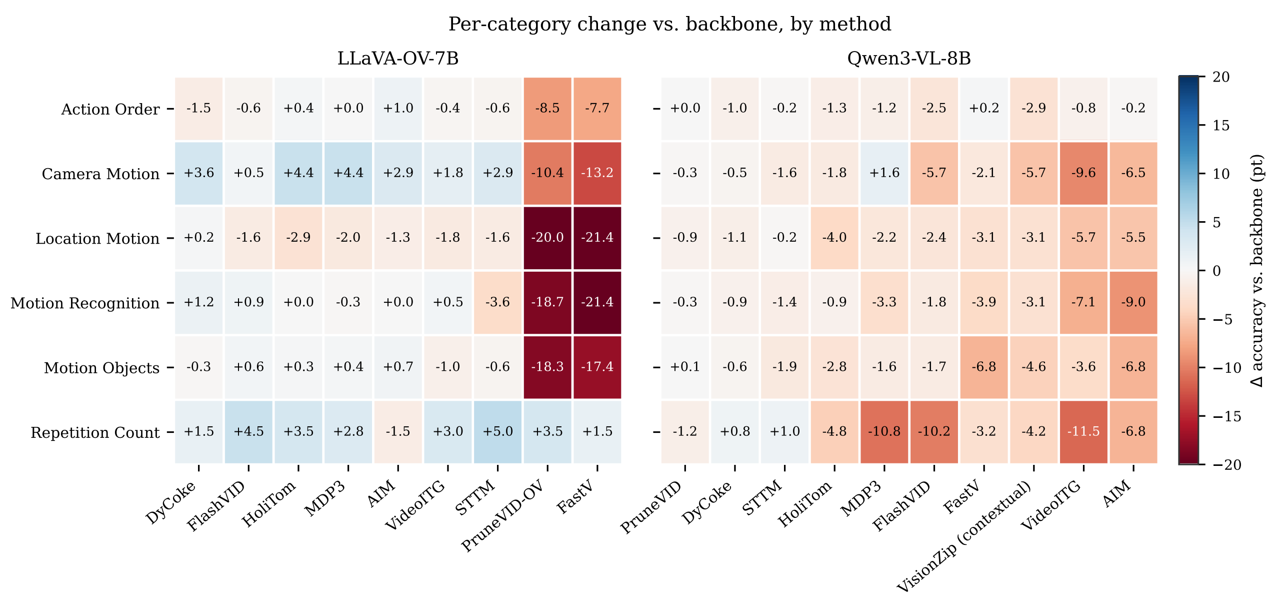

Figure 3: Per-category change vs. backbone, by method (heatmap). Blue = gain, red = loss.

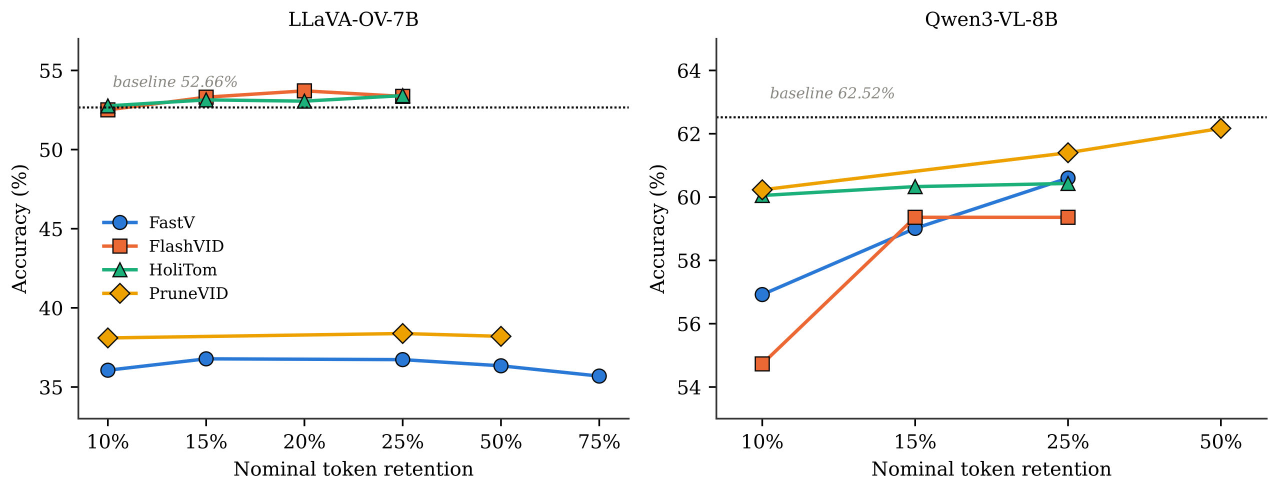

Figure 4: Accuracy against nominal token retention, the four methods with a retention knob (dot + line).

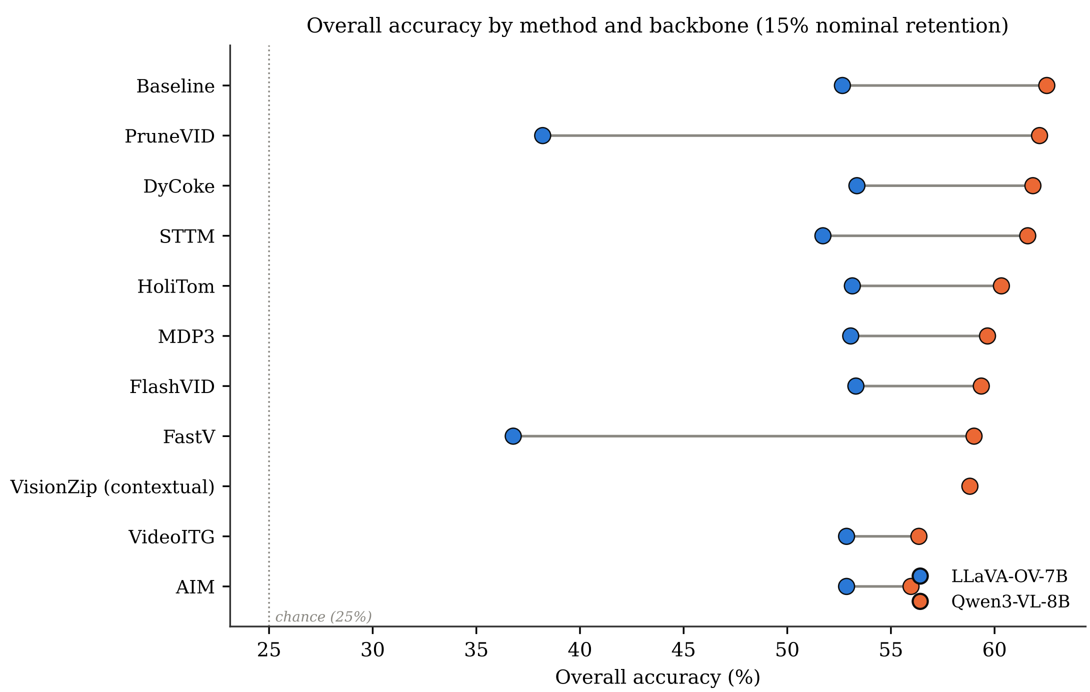

Figure 5: Overall accuracy by method and backbone at 15% nominal retention (Cleveland dot plot). Rows sorted by Qwen3-VL accuracy; the line spans each method's two backbone results.

5 Per-method settings

Table 5 is the mechanism and exact configuration behind every row above.

6 Correct/incorrect vs. baseline

Fixed = wrong under the baseline, correct under the method. *Broke* = correct under the baseline, wrong under the method. Both are McNemar discordant pairs on the 4,018 scoreable items. *Differ*

Table 5: Per-method mechanism, key parameters (as run in this report), and measured visual-token budget. Budgets are read from each method's execution log on Qwen3-VL-8B (11,664 visual input tokens at 32 frames), not from its CLI flag — nominal 15% describes the configured knob, not necessarily what is actually retained. Frame-selection methods have no token budget to measure; they instead reduce the number of frames fed to the model.

Method	Mechanism	Key parameters (this report)	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	`RETAIN_RATIO=0.15 T=0.80 HOLITOM_k=18 HOLITOM_r=0.5`	7.5% (875 tok.)
STTM	quadtree spatio-temporal merge	LLaVA-OV: `thresh=0.85 temporal=0.65 root=1`. Qwen3-VL: `thresh=0.85 temporal=0.65 root=2` — a published setting (`s3_sttm_pub_run`), replacing an earlier off-spec `root=0, temporal=--1.0` cell	9.0% (1,050 tok.) on LLaVA-OV; not re-measured at the corrected Qwen3-VL setting
FastV	in-LLM attention rerank, layer 2	`K=2, R=0.85` (keep 15%). K=2/R=50% (36.34%) and K=2/R=75% (35.69%) were also run and land in the same collapse band (Table 2)	15.0% (1,750 tok.)
FlashVID	pre-LLM temporal-segment merge + inner-LLM compression, layer 28	authors' `flashvid()` package: `retention_ratio=0.15 alpha=0.7 temporal_threshold=0.8 token_selection_method=attn_div pruning_layer=28 llm_retention_ratio=0.1 expansion=1.25 min_segment_num=4`	15.0% (1,750 tok.) pre-LLM; the layer-28 stage compresses surviving visual tokens by a further 90% mid-network, not captured by this single figure
AIM	bipartite merge + PageRank prune	2-step merge schedule compiled in; no CLI retention knob. Qwen3-VL 15%; LLaVA-OV 25% (ships two active steps)	15.0% (1,750 tok.)
VisionZip (contextual)	key-similarity merge in the vision tower	`dominant=54 contextual=10` at 8 frames (native LLaVA-1.5-7B form); Qwen3-VL runs the contextual half only — no CLS token to select the dominant half	15.0% (1,750 tok.)
DyCoke	temporal merge + KV prune, layer 3	`l=3 p=0.7 k=0.7`; no single retention knob, net ∼49%	49.0% (5,716 tok.)
PruneVID	DPC-KNN cluster merge, layer 10	`cluster=0.50 seg=0.25`; LLaVA-OV port has no upstream reference (upstream targets PLLaVA only)	50.0% (5,832 tok.)
MDP3	frame selection (DPP)	`pool=32 select=8` — selects frames, not tokens	n/a (8 of 32 frames)
VideoITG	grounded frame selection	512 sampled → 32 selected	n/a (32 of 512 sampled)
DyTo†	FINCH clustering + ToMe token merge	`temporal_aggregation=spatial_tome_finch_dynamic_all_frms`; own Vicuna-7B backbone	∼3,680 tok. from 100 input frames

† DyTo does not run on either standardized backbone; listed here for completeness (see Table 7).

counts predictions unequal to the baseline's over all 8,052 rows (0 would mean a silent no-op — see `paper.tex` for the verification gate this guards against).

Table 6: Paired-prediction statistics against each backbone's baseline.

Method	Differ	Fixed	Broke	χ^2	Significant
LLaVA-OV-7B					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
Qwen3-VL-8B					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	1,510	133	170	4.28	yes
FlashVID	2,069	157	284	36.00	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
Backbone vs. backbone					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	810	414	127.47	yes

Divergence volume does not predict damage. STTM on LLaVA-OV changes 4,859 predictions — more than FastV's 4,605 — yet loses 0.94 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken).

7 Methods on non-standard backbones

Table 7: Runs on backbones other than the two standardized ones. Not comparable to Tables 3–4; no Δ is given because no matched baseline exists for either row.

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

8 Notable

The backbone swap moves accuracy 9.86 points — more than any token-reduction method moves it, either direction, either backbone, any retention level tested. Rankings do not transfer: PruneVID is the worst LLaVA-OV result (38.20%) and the best Qwen3-VL one (-0.35); Figure 5 shows the crossing lines directly. FastV and PruneVID-OV collapse specifically on LLaVA-OV (37–38%, > 14 points below baseline, significant), while every other LLaVA-OV method is statistically indistinguishable from the backbone; FastV's own published range does not save it (36.3–36.4% at 50% and 75% keep, Table 2). Qwen3-VL inverts the pattern: every method loses, eight of ten losses are significant, but none approaches the LLaVA-OV collapse — the largest Qwen3-VL loss (AIM, -6.55) is smaller than the smallest LLaVA-OV collapse.

Only three methods qualify as having an authors'-validated retention sweep (Table 2); the other five with any retention-like parameter do not. **PruneVID** publishes exactly one ratio. **DyCoke** has no single retention knob, only one fixed three-parameter recipe. **AIM** has no knob at all on LLaVA-OV and an unpublished, self-ported one on Qwen3-VL. **STTM** and **VisionZip** are configured per-benchmark or as fixed token budgets, not a sweepable percentage.